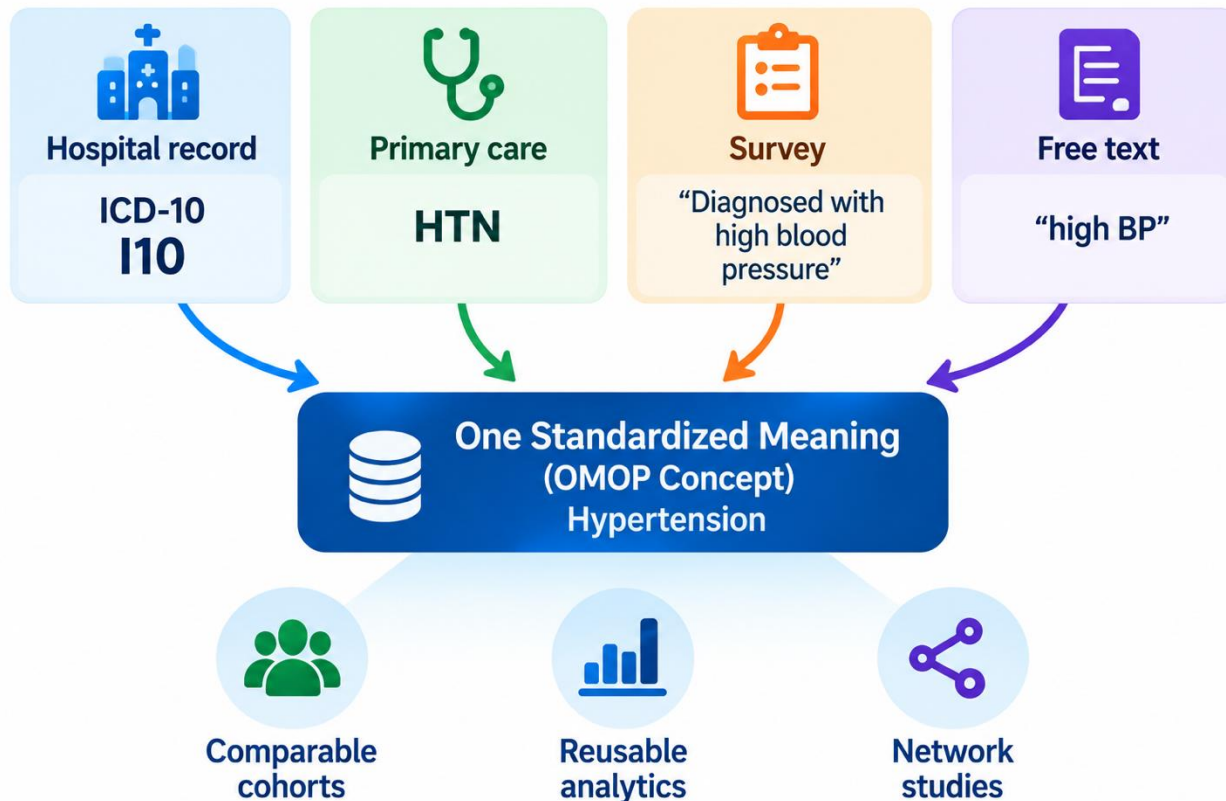




Introduction to OMOP Standardized Vocabularies

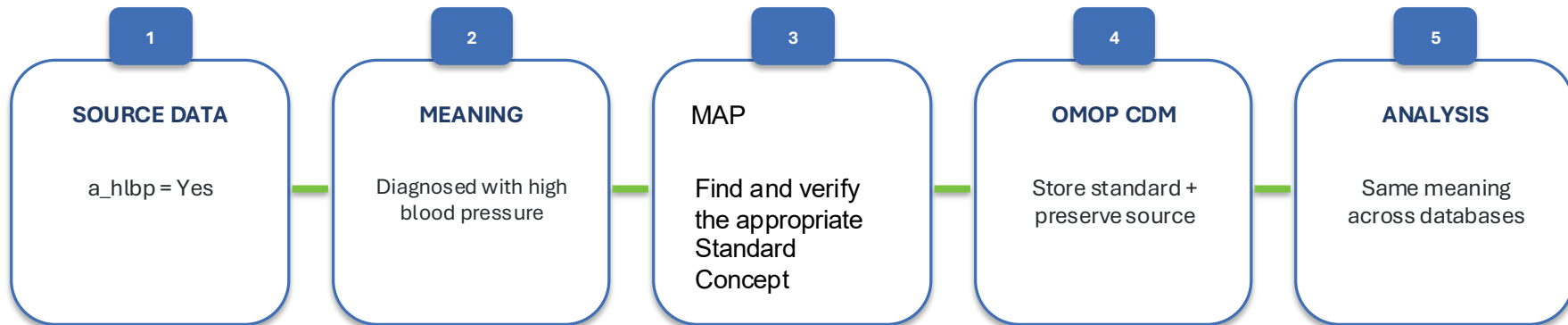
source data: standard concepts: reusable analysis

The Same Clinical Meaning, Many Different Source Representations



From Source Data to Reusable Evidence

OPENING



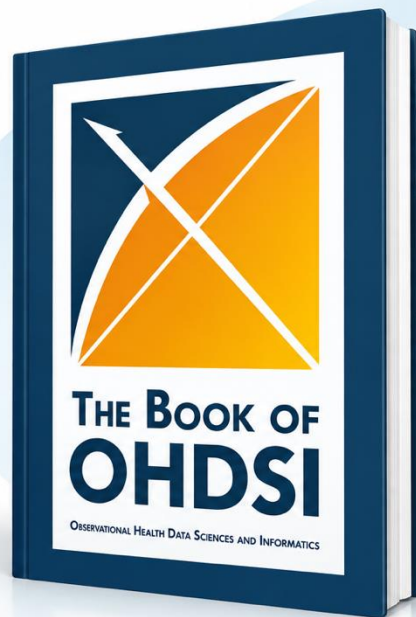
By the end, you should be able to explain why standardization matters, inspect concepts in Athena, and make a defensible simple mapping.

Why Vocabularies Are Foundational to OMOP

“

“...are a **foundational part** of the OHDSI research network, and an integral part of the Common Data Model (CDM). They allow **standardization** of methods, definitions and results by defining the content of the data, paving the way for **true remote** (behind the firewall) **network research and analytics**.”

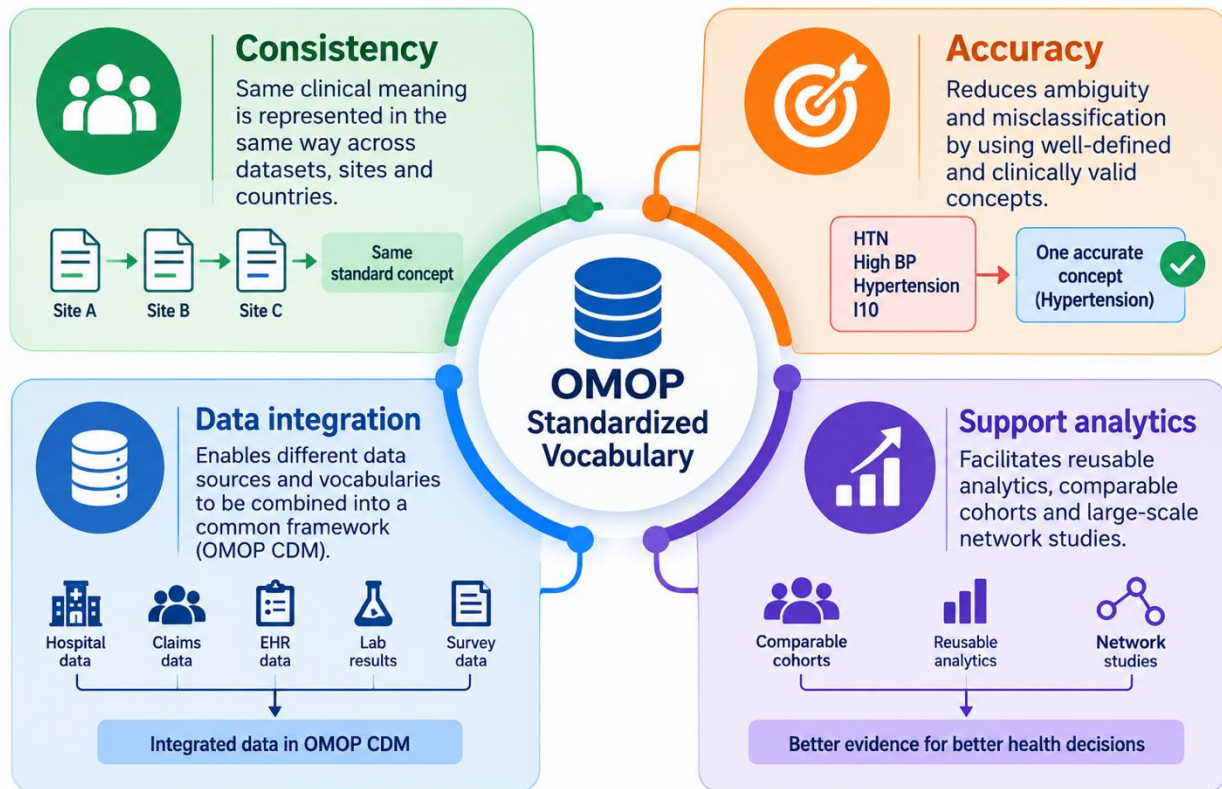
— The Book of OHDSI,
Chapter 5: Standardized Vocabularies



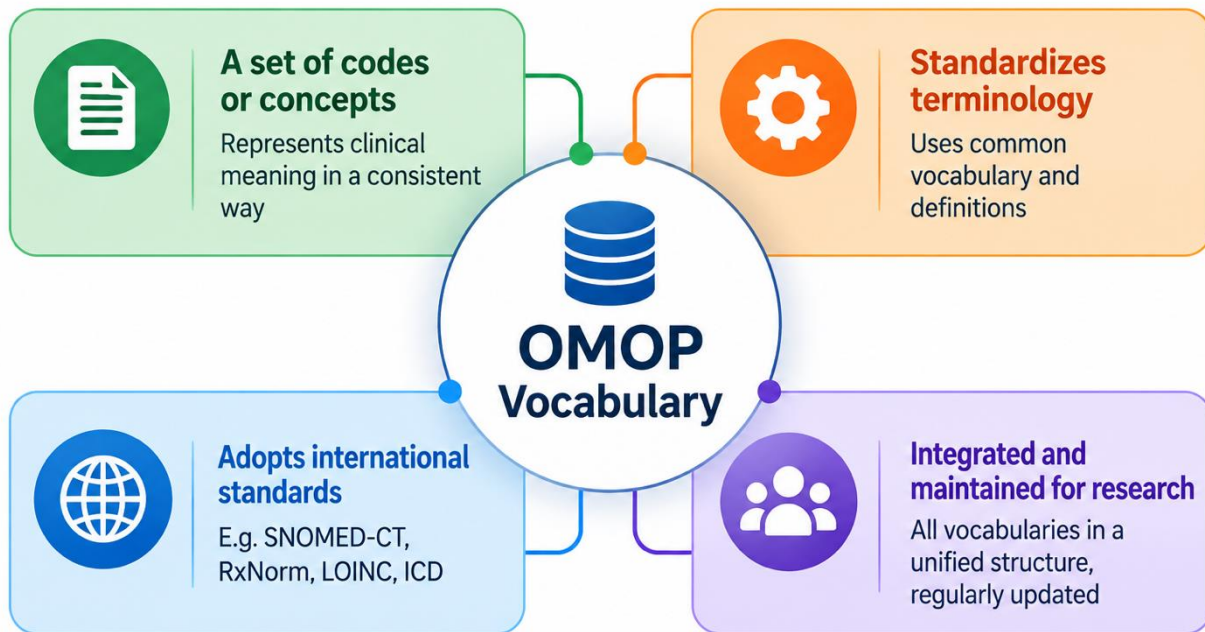
Standardized vocabularies give data a **common meaning**, enabling consistent, accurate and reproducible research.



Why Standardize Terminology?



What Is the OMOP Standardized Vocabulary?



Common Vocabularies You Will Encounter in OMOP



SNOMED-CT

A comprehensive clinical terminology developed for clinical care and research. Widely used to represent clinical findings, diagnoses, procedures and more.



LOINC

A database and universal standard for identifying medical laboratory observations. Also covers survey and questionnaire items, clinical scales and imaging codes.



RxNorm

A standardized terminology that captures medications available in the United States, including clinical drugs, branded drugs and ingredients.



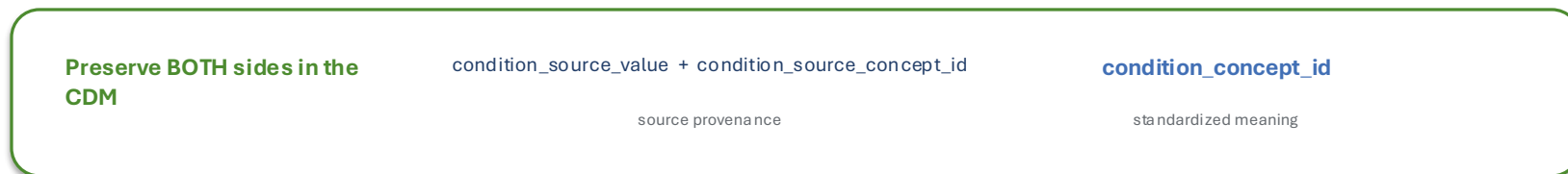
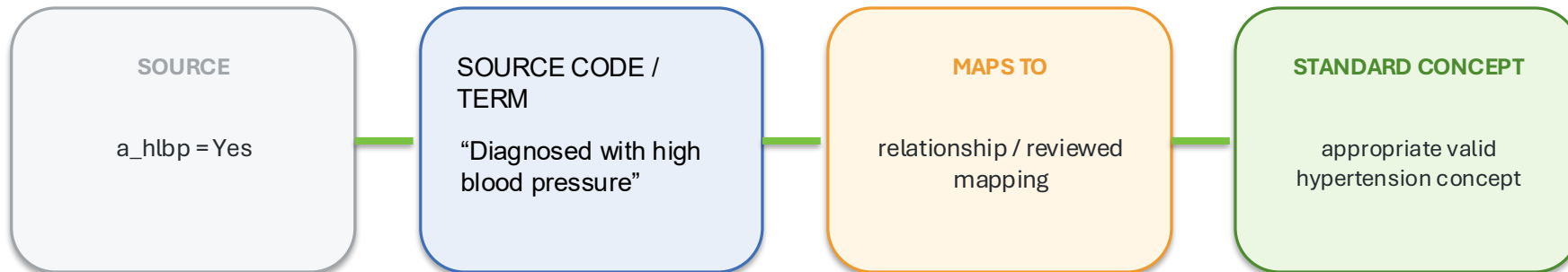
ICD-9 / ICD-10

Classification systems designed for mortality and morbidity reporting, such as public health surveillance and administrative reporting.

The Building Blocks of an OMOP Concept

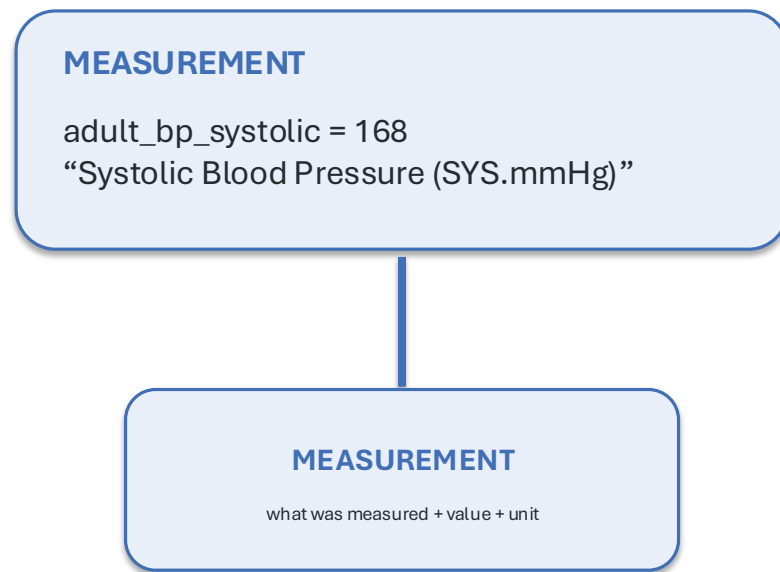
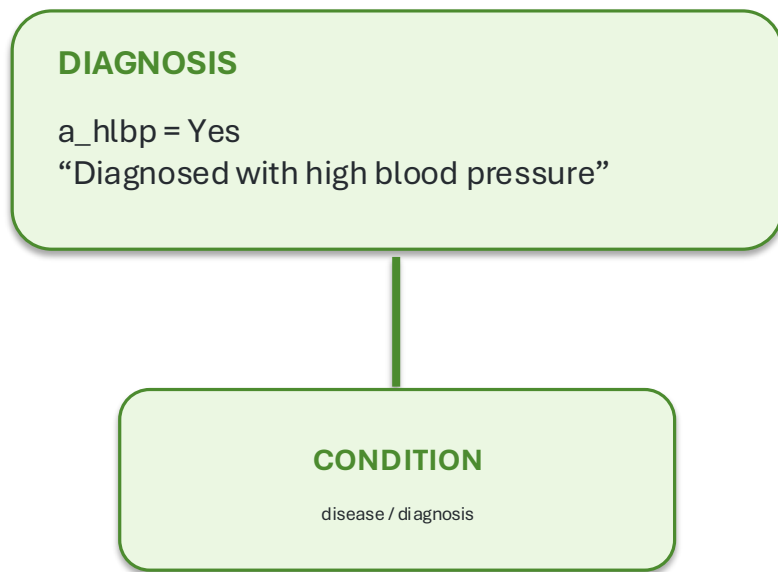
- **DOMAIN:** the broad clinical category a concept belongs to (Condition, Measurement, Drug, Device, Observation)
- **CONCEPT:** a single clinical meaning, identified by a Concept ID
- **CONCEPT CLASS:** the level of detail a concept sits at within its own vocabulary
- **VOCABULARY:** the terminology the concept came from (SNOMED CT, LOINC, RxNorm, ICD-10)
- **STANDARD CONCEPT:** whether the concept may be used in analysis, or must first be mapped
- **VALIDITY:** whether the concept is still current, or has been deprecated

How mapping works: source → Standard



MAPPING RULE Never add meaning the source did not provide.

Same topic ≠ same OMOP concept or domain



Vocabulary mapping is semantic work, not just string matching.

MAPPING RULE Same topic does not mean same concept.

Meet Athena: search, inspect, verify

DEMONSTRATION

Athena is the OHDSI Vocabularies Repository. Search by term or source code, open a candidate concept, then verify it before mapping.

The screenshot shows the Athena web application interface. At the top is a green header with the Athena logo and navigation links: SEARCH, DOWNLOAD, and a user profile for David Amadi. Below the header is a search bar with the text 'aspirin' and a 'Search' button. Under the search bar, there are two instructions: '1. Usage of quotation marks forces an exact-match search' and '2. In case of a typo, or if there is a similar spelling of the word, the most similar result will be presented'. Below the search bar is a section titled 'Explore domains' with six tiles: Drugs (6,034,243), Conditions (729,078), Procedures (817,314), Devices (556,702), Observations (1,033,599), and Measurements (594,467).

Five checks before accepting a mapping

1

Meaning

Does it mean what the source means?

2

Domain

Correct clinical domain?

3

Standard

STANDARD_CONCEPT = S?

4

Validity

Current / not invalid?

5

Specificity

Am I adding unsupported detail?

Search by term or source code → open a candidate → inspect the concept record

Do not memorize Concept IDs — verify the vocabulary release used for training.

Standard representation of vocabulary concepts

CONCEPT_ID	313217	← Primary key
CONCEPT_NAME	Atrial fibrillation	← English description
DOMAIN_ID	Condition	← Domain
VOCABULARY_ID	SNOMED	← Vocabulary
CONCEPT_CLASS_ID	Clinical Finding	← Class in vocabulary
STANDARD_CONCEPT	S	← Standard, Source of Classification
CONCEPT_CODE	49436004	← Code in vocabulary
VALID_START_DATE	01-Jan-1970	← Valid during time interval
VALID_END_DATE	31-Dec-2099	
INVALID_REASON		

Source: The Book of OHDSI

The same record, traced end to end:

ICD9CM 427.31 (concept_id 44821957 · non-standard) — “Maps to” → SNOMED 49436004 (concept_id 313217 · standard · domain_id = Condition) → CONDITION_OCCURRENCE.condition_concept_id

The source code is kept in condition_source_value and condition_source_concept_id. The domain — not the column it came from — chose the table.

Your turn: map six source variables

Work in pairs. Complete A, B and C, then attempt at least one of D, E or F. Fill the four right-hand columns yourself and briefly justify your choice.

	SOURCE VARIABLE	DOMAIN	VOCABULARY	CONCEPT ID	MAPPING RATIONALE
A	a_hlast	_____	_____	_____ -	_____
B	adult_bp_systolic	_____	_____	_____ -	_____
C	a_hldia	_____	_____	_____ -	_____
D	a_emodep	_____	_____	_____ -	_____
E	fluoxetine	_____	_____	_____ -	_____
F	Cesarean section	_____	_____	_____ -	_____

For each: what does the source mean? → which domain? → candidate concept → Standard? → Valid? Never add specificity the source does not support.

When Athena is not enough: Usagi

For local terms or larger mapping jobs,
Usagi ranks candidate concepts for
human review.

- 1 LOCAL SOURCE TERMS**
e.g., “heart problems”
- 2 USAGI**
ranks candidate Standard Concepts
- 3 HUMAN REVIEW**
clinical + source-system knowledge
- 4 APPROVED MAPPING**
document the rationale and use in ETL

Overview table

Selected mapping

Search facility

The screenshot shows the Usagi application interface. The top section, labeled 'Overview table', displays a table with columns for Status, Source code, Source term, Frequency, Dutch term, Match score, Concept ID, Concept name, Domain, Concept class, Vocabulary, Concept code, Standard concept, Parents, Children, and Comment. The middle section, labeled 'Selected mapping', shows a table with columns for Source code, Source term, Frequency, and Dutch term, and a table below it with columns for Concept ID, Concept name, Domain, Concept class, and Vocabulary. The bottom section, labeled 'Search facility', includes a search bar, filters, and a results table with columns for Score, Terms, Concept ID, Concept name, Domain, Concept class, Vocabulary, Concept code, Standard concept, Parents, and Children.

Usagi suggests. The mapper decides.

If no defensible Standard Concept exists, preserve the source rather than forcing a close-enough mapping.

Usagi supports candidate mapping and review; local mapping governance should be documented explicitly.

Mapping in Practice: Challenges and Lessons Learned

■ **Vocabulary mapping is rarely a clean lookup. Context, expertise and documentation decide the outcome.**

■ **Context & Localization**

■ Population-health and programme concepts often have no neat Standard Concept.

■ **Domain Expertise**

■ Good mappings need someone who knows both the source context and the clinical terminology.

■ **Documentation & Governance**

■ Record the rationale, and revisit it at every vocabulary release.



Thank you for your attention!